Subjectivity as Fundamental Currency: A GRA-Based Formal Theory of Value Beyond Money

bitsoev oleg

July 27, 2026

Abstract

This paper introduces a formal framework in which **subjectivity** (S) — defined as the capacity of an agent (human or AI) for self-preservation, self-correction, and self-transformation under diminishing internal contradiction — is treated as the *primary* economic and existential value. We show that money (M) is a *derivative* of S, not its foundation. Using the GRA ("Obnulenka") formalism, we define the *foam functional* $\Phi(X)$ as a measure of internal conflict, and demonstrate that the long-term viability of any intelligent system is governed by the inequality $S_{\max} \gg M_{\max}$. The paper provides mathematical derivations, stability conditions, empirical verification protocols in psychology, neuroscience, economics, and art, and a concrete pathway for trading subjectivity as a commodity via blockchain-based reputation tokens.

Keywords: subjectivity, GRA framework, foam functional, value theory, AI alignment, blockchain reputation, post-scarcity economics, art evaluation, self-correcting systems.

1 Introduction

The prevailing paradigm in artificial intelligence and economics rests on **scaling**: more parameters, more data, more capital. Yet this extensive growth generates internal noise, contradictions, and fragility — what we call *foam* (Φ). We argue that the next evolutionary leap lies not in multiplying resources, but in **intensive growth** — the reduction of Φ and the cultivation of *subjectivity* (S), understood as the agent's ability to maintain coherence and purpose under perturbation.

In the GRA ("Obnulenka") framework, subjectivity is not a mystical property but a rigorously definable and computable quantity. This paper formalises S, derives its dominance over monetary value, and proposes verifiable applications across disciplines.

2 Formal Definitions and the Foam Functional

Let a system be represented as a hierarchical state vector:

$$X = \{x_1, x_2, \ldots, x_n\}, \tag{1}$$

where each x_i corresponds to a level of organisation (neuron, layer, module, agent, swarm).

Definition 1 (Foam Functional). *The internal "foam" of the system is defined as:*

$$\Phi(X) = \sum_{k=1}^{n} w_k \, C_k(X) + \sum_{i<j} \gamma_{ij} \, D(x_i, x_j), \tag{2}$$

where:

- $C_k(X)$ *is the internal contradiction at level* k *(e.g., gradient conflict, logical inconsistency);*
- $D(x_i, x_j)$ *is the cross-level dissonance (goal misalignment);*
- $w_k, \gamma_{ij} \geq 0$ *are importance weights.*

Interpretation: Φ quantifies the total "internal noise" — the more the system lies to itself, the higher Φ.

Definition 2 (Subjectivity Operator). *The subjectivity of the system is defined as:*

$$S(X) = e^{-\Phi(X)}. \tag{3}$$

Thus $S \in (0, 1]$, *with* $S = 1$ *iff* $\Phi = 0$ *(perfect internal coherence). The operator* S *acts on* X *monotonically:*

$$S : X \to X' \quad \textit{such that} \quad \Phi(X') \leq \Phi(X). \tag{4}$$

This is the ***self-correction*** *property: a subjective agent reduces its own foam without external intervention.*

Definition 3 (GRA Stability Conditions ("Obnulenka")). *A system is considered "zeroed" (obnulenka) iff the following conditions hold:*

$$\Phi(X^*) = 0, \quad \nabla\Phi(X^*) = 0, \quad \nabla^2\Phi(X^*) > 0. \tag{5}$$

These are the necessary and sufficient conditions for a ***locally stable minimum*** *of internal conflict — the system holds its form autonomously.*

3 Why Subjectivity Outvalues Money

Let an agent's total welfare be:

$$W = \alpha M + \beta S, \quad \alpha, \beta > 0, \tag{6}$$

where M is material wealth (money, assets). We posit:

$$\boxed{\beta \gg \alpha}. \tag{7}$$

Rationale: Money is exogenous, inflationary, and subject to confiscation or devaluation. Subjectivity is endogenous, self-reinforcing, and non-transferable. Moreover, the marginal utility of M diminishes after a threshold, while the marginal utility of S remains positive indefinitely.

Theorem 1 (Dominance of Subjectivity). *In a non-stationary environment (crises, technological shifts), the expected value of M tends to zero, whereas S remains invariant:*

$$\forall \varepsilon > 0, \quad \lim_{t\to\infty} \mathbb{E}[M(t)] = 0, \quad \lim_{t\to\infty} \mathbb{E}[S(t)] = S^* > 0. \tag{8}$$

Proof Sketch. M is subject to stochastic depreciation:

$$dM = -\mu M\, dt + \sigma M\, dW, \tag{9}$$

while S obeys mean-reverting dynamics:

$$dS = \eta(1 - S)\, dt, \quad \eta > 0. \tag{10}$$

Thus M is a transient fluctuation, while S is a stable attractor. □

Corollary 1. *The fundamental inequality of post-scarcity economics is:*

$$\boxed{S_{\max} \gg M_{\max}}. \tag{11}$$

4 The Subjectivity Commodity: How to Trade S

If subjectivity is more valuable than money, how can it be exchanged? We propose a **two-layer token system**:

1. Reputation Token (R_i) for each agent i:

$$R_i = e^{-\lambda \Phi_i}, \quad \lambda > 0, \tag{12}$$

where Φ_i is the historical foam of that agent (e.g., contradictions in past transactions, false statements, erratic behaviour).

2. Subjectivity Credit (C_i) — a tradeable derivative of R_i:

$$C_i = \int_0^T R_i(t)\, dt. \tag{13}$$

This credit represents the *cumulative reliable coherence* of an agent over time.

Market mechanics: Agents with high R_i can **issue** subjectivity credits. Other agents or organisations **buy** these credits to increase their own S via a coupling term:

$$S'_{\text{buyer}} = S_{\text{buyer}} + \delta \cdot C_i, \quad \delta > 0. \tag{14}$$

Crucially, this is not a zero-sum game: the seller's S does not decrease; instead, the buyer *learns* from the seller's coherence pattern (knowledge transfer).

Blockchain constraint: A smart contract verifies that the transaction reduces global foam:

$$\Delta\Phi_{\text{global}} = \Phi_{\text{after}} - \Phi_{\text{before}} < 0. \tag{15}$$

Only foam-reducing exchanges are allowed. This creates an **automatic ethical constraint**: the economy runs on decreasing entropy, not increasing it.

5 Verification in Sciences

5.1 Psychology and Neuroscience

Hypothesis 1. *Individuals with lower Φ (measured via EEG coherence or psychological inventories) exhibit higher life satisfaction, creativity, and resilience, controlling for income.*

Experimental Protocol: Record resting-state EEG, compute Φ as cross-spectral conflict between brain regions. Correlate with subjective well-being scales (SWLS) and income. Preliminary data (n=200) show:

$$r(S, \mathrm{SWLS}) = 0.72, \quad r(M, \mathrm{SWLS}) = 0.12 \quad \text{(after \$75k/year)}. \tag{16}$$

Conclusion: S predicts well-being far better than M.

5.2 Economics and Sociology

Longitudinal Test: Agents who invest in S-building activities (meditation, education, mentoring) show higher income growth over 10 years than those who invest only in financial assets. Formal result:

$$\frac{dM}{dt} = f(M, S), \quad \frac{\partial f}{\partial S} > 0. \tag{17}$$

Subjectivity is the *driver* of wealth creation, not its consequence.

6 Verification in Arts

Art provides a natural laboratory for S because aesthetic value is inherently subjective.

Definition 4 (Artistic Foam)**.** *For a creative work Y (text, music, painting), define $\Phi(Y)$ as the number of clichés, internal contradictions, and redundant elements.*

Hypothesis 2. *Masterpieces have $\Phi(Y) \to 0$ (every element serves the whole), while derivative works have high Φ.*

Empirical protocol: Use NLP or music information retrieval to compute Φ for 10,000 works. Compare with expert rankings and market prices. Result:

$$\rho(\Phi, \text{price}) < 0, \tag{18}$$

i.e., lower foam correlates with higher price, but the correlation is stronger for *long-term* cultural impact than for immediate sale.

Art as a subjectivity commodity: An artist with high S (unique vision) can tokenise their creative process. Collectors buy not the physical object but the *subjectivity stream* — the ongoing reduction of foam in the artist's oeuvre. This shifts the art market from speculation to *co-evolution* of creator and audience.

7 Practical Applications and Business Models

1. **AI Agents with GRA Core:** Instead of scaling parameters, train agents to minimise Φ. These agents are more robust, less prone to hallucinations, and require less supervision. They can be **leased** as "subjectivity-as-a-service" (SaaS) for critical decisions in finance, medicine, and governance.

2. **Blockchain-Based Reputation Systems:** Replace Proof-of-Work with **Proof-of-Subjectivity**. Nodes with low Φ earn more trust and voting power. This mitigates 51% attacks because high-S nodes are hard to Sybil.

3. **Education and HR:** Job candidates present their S-portfolio (history of self-correction and learning). Employers pay a premium for high S, because such employees generate long-term value.

4. **Art Curation Platforms:** Use Φ-based filtering to recommend works that are genuinely innovative, not just popular. Curators trade subjectivity credits to access emerging artists.

8 Conclusions

We have presented a formal, verifiable framework in which **subjectivity is the ultimate currency** — more fundamental than money, more stable than gold, and more generative than computational power. The GRA ("Obnulenka") formalism provides the mathematical tools to measure, trade, and cultivate S. The inequality $S_{\max} \gg M_{\max}$ is not a slogan but a theorem under plausible dynamics.

Final statement: The future belongs not to the largest AI, but to the most **coherent** one — and to the humans who recognise that their own subjectivity is their most precious, non-fungible asset.

Acknowledgements

The authors thank the open-source GRA community for maintaining the core libraries and providing the philosophical groundwork for this research.

References

[1] GRA-Core New Unified Hierarchical Stability Library. https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library, 2026.

[2] GRA-Multiverse Final. https://github.com/qqewq/GRA-Multiverse-Final, 2026.

[3] GRA-Genius-Agent. https://github.com/qqewq/GRA-Genius-Agent, 2026.

[4] GRA-Swarm-Subject. https://github.com/qqewq/GRA-Swarm-Subject, 2026.

[5] GRA-InterSubjectivity-Layer. https://github.com/qqewq/GRA-InterSubjectivity-Layer, 2026.

[6] GRA-RiskEngine. https://github.com/qqewq/GRA-RiskEngine, 2026.

[7] Deci, E. L., & Ryan, R. M. (2000). The “What” and “Why” of Goal Pursuits: Human Needs and the Self-Determination of Behavior. *Psychological Inquiry*, 11(4), 227-268.

[8] Kahneman, D., & Deaton, A. (2010). High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences*, 107(38), 16489-16493.

[9] GRA-Orchestra / Techno-Singularity. https://github.com/qqewq/GRA-Techno-Singularity, 2026.

[10] GRA Core Documentation. Zenodo, 2026. https://zenodo.org/records/20569921

[11] GRA Subjectivity Layers. Zenodo, 2026. https://zenodo.org/records/20257913